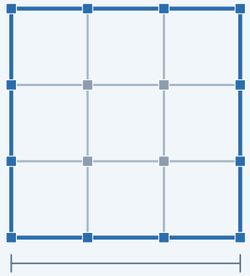
SeisFrame automated structural design

Explore options. Test performance. Make engineering-guided changes.

Changed members Higher grade

01 Define the building

Building layout



Bays · spans · stories

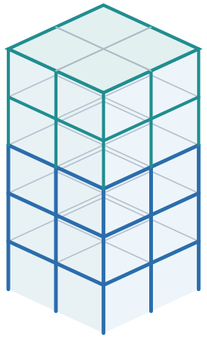


Gravity + earthquake

Material & detailing limits

02 Explore design options

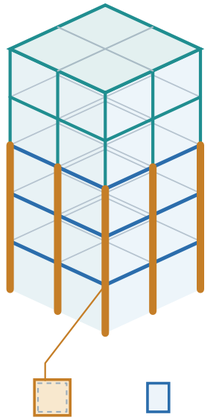
A Starting option



Column Beam

Column	24 × 24
Beam	18 × 24
Concrete	5 ksi

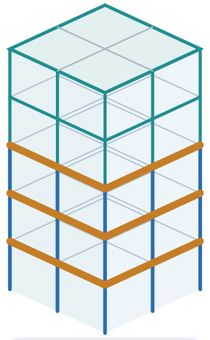
B Wider columns



Column Beam

Column	**30 × 30**
Beam	18 × 24
Concrete	5 ksi

C Deeper beams

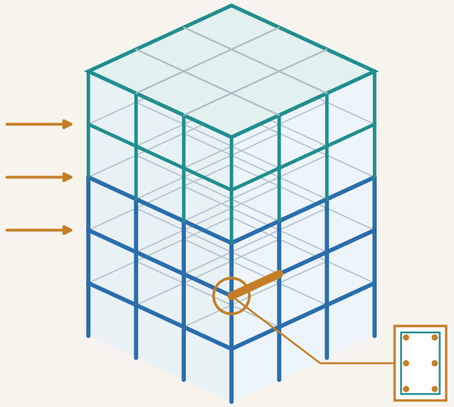


+ stronger concrete

Column Beam

Column	24 × 24
Beam	**18 × 30**
Concrete	**6 ksi**

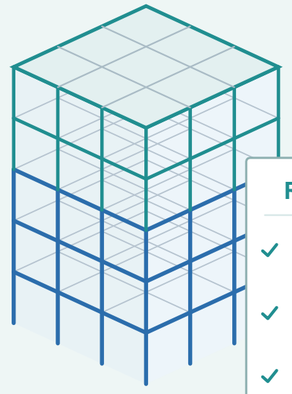
03 Test each generation



Locate governing weaknesses

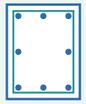
- Member strength
- Joints & detailing
- Sway & stability

04 Verify & save



RECORD

- ✓ Sections
- ✓ Rebar
- ✓ Checks



Reinforcement

selected + checked

Final verification + design record

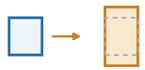
GENETIC ALGORITHM LOOP

Next generation



Updated design candidates

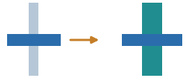
Mutate with engineering guidance



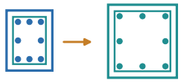
Member size



Concrete grade



Column balance



Joint & bar fit



Sway control

Select promising designs



Compare checks + preserve diversity

Test → select → mutate → next generation → test again